

Employability Skills Inventory

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About this employability skills inventory

Throughout this course, you've advanced your knowledge, skills, and fluency in cybersecurity. You've researched job roles and explored certifications that you might want to work toward.

You've also practiced employability skills that employers consider essential. Although you worked on these skills in the context of cybersecurity, remember that these skills are transferable; they will be valuable in any career you pursue.

This document summarizes the employability skills you strengthened and give examples of activities you engaged in to practice them. Use this inventory's wording in your resume, in job interviews, and at networking events.

Module 1: Data and Privacy

- Critical thinking
 - Explain why organizations need to invest in cybersecurity
 - Identify cybersecurity threats
 - Classify tasks using the key elements of cybersecurity
 - Categorize types of data
 - Identify privacy best practices
 - Decrypt encrypted text using a cipher key
 - Order the steps for encryption
 - List the reasons that encryption is so important for organizations
 - Apply best practices of encryption
- Growth mindset
 - Explain your interest in a career in cybersecurity
 - Describe the required skills to have a successful career in data privacy
 - Describe the required skills to succeed as a data security analyst
 - Summarize the purpose, objectives, and benefits of the EC-Council Certified Encryption Specialist (ECES) program
- Analytical thinking
 - Distinguish between the consequences of a data privacy breach
 - Apply the CIA triad to real-world situations
 - Sort controls by type
 - Classify characteristics of encryption as symmetric or asymmetric
- Written communication, documentation
 - Develop a backup plan

Module 2: Governance, Risk, and Compliance

- Growth mindset
 - Reflect on the application of organizational policies
 - Describe methods for securing data, systems, and networks
 - List technical and workplace skills that employers look for when filling open security compliance analyst positions.
- Critical thinking
 - Differentiate between policies, standards, guidelines, and procedures
 - List reasons that a robust cybersecurity compliance program is critical for protecting sensitive customer data.
 - Apply federal laws to real-world situations
- Critical thinking, written communication
 - Assess organizational risk
- Problem solving, documentation
 - Create an administrative control plan for compliance

Module 3: Threats and Vulnerabilities

- Critical thinking
 - List types of malware that threaten unsecured devices
 - Differentiate between different types of malware
 - List signs of a social engineering attempt
 - Identify ways that a seemingly innocent visitor can pose a physical security threat to an organization
- Critical thinking, systems thinking
 - Analyze the impact of cybersecurity threats on systems and processes
- Problem solving, analytical thinking, learning agility
 - Mitigate malware threats using Malwarebytes
- Research
 - Describe the skills and knowledge required to work as a malware analyst
- Attention to detail
 - Apply the characteristics of phishing to emails
- Written communication, documentation
 - Develop a physical security control plan
- Growth mindset, research
 - Describe the required skills to succeed as a physical security service technician

Module 4: Vulnerability Management

- Critical thinking
 - Identify the steps of the threat intelligence lifecycle
- Analytical thinking
 - Analyze threat intelligence
 - Analyze a STIX expression
 - Categorize vulnerabilities by severity
 - Identify the four phases of penetration testing in a realistic scenario
- Growth mindset, research
 - Describe the required skills to succeed as a vulnerability assessment analyst or manager
 - List the benefits of completing the CompTIA PenTest+ certification
- Critical thinking
 - Evaluate the current state of organizational security
- Problem solving, analytical thinking, learning agility
 - Analyze a web application's vulnerabilities using OWASP ZAP

Module 5: System Security

- Analytical thinking, written communication
 - Explain why system security is as important as the physical security of a home or office building
- Resourcefulness, written communication
 - Create a plan to ensure the security of your firmware
- Growth mindset, written communication
 - List requirements for becoming a firmware engineer
 - List key information about the Red Hat Certified System Administrator (RHCSA) certification
- Critical thinking
 - Determine the appropriate server
 - Classify operating systems
 - Describe system hardening
- Problem solving, learning agility
 - Secure a host operating system
- Critical thinking, documentation
 - Create a plan to secure an operating system

Module 6: Network Security

- Analytical thinking, written communication
 - List network security measures that organizations should implement
 - Explain how implementing network authentication and authorization can help organizations maintain the confidentiality and security of client data
- Analytical thinking
 - Classify application and service attacks by type
 - Classify wireless attacks by type
 - Classify strategies for securing networks
 - Design a secure network
- Critical thinking
 - Describe encryption protocols
 - Describe the purpose of network security devices
 - Identify advantages of devices used in protecting network security
- Growth mindset, written communication
 - List technical skills needed to manage an organization's network
 - Explore the CompTIA Network+ certification and your career path
- Research, written communication
 - Summarize the typical tasks that an identity and access management analyst performs
- Creative thinking
 - Differentiate between network access controls

Module 7: Cloud Computing and Virtualization

- Analytical thinking, written communication
 - Explain the benefits of cloud computing for an organization
- Critical thinking
 - Identify benefits of virtualization
 - Identify key features of external and internal virtualization
 - Identify the appropriate cloud service model based on its characteristics
- Problem solving, learning agility
 - Create a virtual machine
- Growth mindset, research
 - List courses that can help prepare you for a career as a virtualization engineer
 - Describe the skills required to succeed in a cloud computing career
- Attention to detail
 - Apply the appropriate cloud deployment model given an organization's needs
- Analytical thinking, documentation
 - Describe when to implement different cloud deployment models

Module 8: Securing Cloud Infrastructure

- Analytical thinking, written communication
 - List some of the threats unique to cloud computing
- Analytical thinking, attention to detail
 - Distinguish between types of mitigation for DoS and DDoS attacks
 - Identify infrastructure visibility challenges
- Growth mindset, research
 - Reflect on the steps you would need to take to become a cloud security analyst
- Critical thinking
 - Describe shadow IT
- Problem solving, Attention to detail
 - Apply identity access management to real-world situations
- Problem solving, analytical thinking, learning agility
 - Create a secure virtual machine using Microsoft Azure
- Critical thinking, written communication
 - List the effects of a cloud service outage on everyday life
- Attention to detail
 - Apply cloud infrastructure and recovery plans
- Growth mindset, written communication
 - Reflect on your interest in various skills tested as part of the CompTIA Cloud+ certification
- Adaptability and resilience, analytical thinking
 - Apply security strategy principles to cloud applications
- Learning agility, written communication
 - Summarize key concepts for securing cloud applications

Module 9: Security Operations

- Analytical thinking, written communication
 - Describe potential limitations or challenges that SOC teams might face when using tools such as IBM QRadar and AI analysis
 - Explain the importance of strong communication skills for cybersecurity professionals
- Attention to detail
 - Distinguish between standard security models
- Attention to detail, learning agility
 - Identify methods of preparation, planning, and prevention
 - Identify responsibilities for monitoring, detection, and response
 - Identify responsibilities for recovery, refinement, and compliance
- Attention to detail, creative thinking
 - Assign SOC roles and tasks
- Growth mindset, research
 - List common requirements for SOC analyst jobs
- Critical thinking
 - Identify types of security awareness training
- Creative thinking
 - Identify the components of security awareness training

Module 10: Security Monitoring

- Analytical thinking, written communication
 - List ways that endpoint detection and response (EDR) enhances the security of an organization's network and how it differs from traditional endpoint security tools
- Critical thinking
 - Identify network diagnostic and performance monitoring tools
 - Identify the benefits of EDR tools
 - Identify important features to search for in an EDR tool
 - Identify the benefits of SIEM tools
- Problem solving, learning agility
 - Perform network reconnaissance
- Growth mindset, written communication
 - Explore the CompTIA CySA+ certification and your career path
- Attention to detail, creative thinking
 - Apply principles of endpoint management and monitoring to use an EDR tool
- Critical thinking, written communication
 - Explain how SIEMs help teams prioritize alerts and improve the security of their data
- Problem solving, learning agility
 - Perform threat investigation using Splunk Enterprise
- Growth mindset, research
 - List common requirements for threat hunter jobs

Module 11: Incident Response

- Critical thinking, written communication
 - List the most critical elements of a well-prepared incident response plan
 - Apply an appropriate incident response framework given an organization's needs
 - Identify the four phases of an incident response
- Critical thinking
 - Identify roles on the incident response team
 - Identify participants outside of the incident response team with responsibilities in the event of an incident
 - Identify specific concerns related to each incident response phase that an incident response plan needs to address
 - Apply the Cyber Kill Chain framework
 - Apply the MITRE ATT&CK framework
- Growth mindset, research
 - List common requirements for incident responder positions
 - List the EC-Council's nine stages of incident handling
- Analytical thinking
 - Distinguish among key areas of intrusion analysis
- Problem solving, learning agility
 - Respond to a network attack

Module 12: Digital System Forensics

- Systems thinking, written communication
 - Explain how various skills and knowledge contribute to the overall integrity of the digital forensics process
- Analytical thinking
 - Analyze a cyberattack through digital forensics
- Growth mindset, written communication
 - Describe the required skills to succeed as a digital forensics investigator
- Learning agility
 - Apply the four phases of digital forensics to a scenario
- Critical thinking, written communication
 - Describe applications of problem-solving skills to digital forensics investigations
- Attention to detail
 - Identify the purpose of digital forensic tools
- Problem solving, learning agility
 - Analyze digital forensics evidence using FTK Imager and Autopsy

Module 13: Emerging Threats and the Future of Cybersecurity Technologies

- Growth mindset, written communication
 - List emerging technologies of which you were already aware
 - Examine the skills that you know best and the skills that you'd like to improve
 - Identify the certifications that you'd like to pursue
 - Identify the jobs that appeal to you the most and why
- Analytical thinking, written communication
 - Summarize emerging cybersecurity threats
 - Describe the future of cybersecurity technologies
 - Describe the impact of emerging technology in the field of cybersecurity

Next steps

There are important next steps you can take to further your career and plan a professional future.

Certifications

Certifications are another step you can take to learn more and advance in your career. You explored the following certifications related to cybersecurity. Research these and other certifications that interest you to see if you can commit to completing one:

- **EC-Council certified encryption specialist (ECES):** You explored the ECES certification and careers specializing in encryption.
- **CompTIA PenTest+ certification:** You explored this certification, which is helpful for a career in penetration testing.
- **Red Hat Certified System Administrator (RHCSA):** You explored this certification, which helps with a career in managing and hosting operating systems.
- **CompTIA Network+ certification:** You explored this certification, which helps with a career in network administration and systems administration.
- **CompTIA Cloud+ certification:** You explored this certification, which is helpful for jobs such as cloud engineer, cloud specialist, cloud project manager, network engineer, and cloud security specialist.
- **CompTIA cybersecurity analyst (CySA+) certification:** You explored this certification and the jobs associated with it, including application security analyst, threat intelligence analyst, vulnerability assessment analyst, security architect, and cybersecurity engineer.
- **Certified Incident Handler (CIH) certification:** You explored this certification offered by the EC-Council, which helps cyberprofessionals prepare to manage and respond to cybersecurity incidents.

Careers

You also learned about a variety of cybersecurity careers and career paths. These specialized jobs require additional training, certification, and experience. You can get started on these paths by using your cybersecurity fluency to contribute to the information and systems security in any role or industry you work in. If you are interested in any of the following jobs, use that as your focus for setting goals and planning your future in the field of cybersecurity.

You explored the following content about cybersecurity careers:

- **Cybersecurity career path:** You identified job roles in cybersecurity and learned the typical career progression in cybersecurity.
- **Data privacy career paths:** You identified job roles in data privacy, which involve ensuring compliance with privacy laws and regulations, performing technical work related to privacy, and overseeing various privacy-related tasks.
- **Data security analyst:** You explored the details of this job, which include actively monitoring and protecting data from unauthorized access or breaches.
- **Data, system, and network security:** You explored jobs as a cybersecurity professional, which involves reducing risk to data, systems, and networks.
- **Security compliance analyst:** You explored the details of this job, where you actively ensure adherence to security standards and regulations.
- **Malware analyst:** You explored the details of this job, which involves helping to draft your company's cybersecurity policies, standards, guidelines, and procedures and responding to reported malware incidents, limiting their spread, and helping with recovery efforts.
- **Physical security service technician:** You explored jobs in physical security, where you'll install, support, and troubleshoot physical security controls such as security cameras, biometric scanners, and door access control systems.
- **Vulnerability assessment analyst and vulnerability assessment manager:** You explored the details of these roles. A vulnerability assessment analyst actively identifies and assesses security vulnerabilities in systems, networks, or applications. A vulnerability assessment manager oversees the organization's process and coordination of vulnerability assessments.
- **Firmware engineer:** You explored the details of this job, which involves developing and troubleshooting firmware.
- **Network administrator:** You explored the details of a network administrator career, which manages and maintains a computer network so that users and services can access network resources efficiently and securely.
- **Identity and access management (IAM) analyst:** You explored the details of this job, which involves managing and securing user access to company resources such as computer systems, networks, and applications.
- **Virtualization engineer:** You explored this job, which is responsible for designing, implementing, and managing virtualization solutions that help businesses save money,

improve efficiency, and increase agility. You learned how to gain the necessary skills to succeed.

- **Cloud computing:** You explored the jobs in cloud computing, which involve designing, implementing, and managing cloud-based infrastructure and services for organizations.
- **Cloud security analyst:** You explored this career, where you evaluate potential security risks and develop plans to mitigate them, monitor systems for unauthorized access or breaches, and implement security controls and procedures to protect against threats.
- **SOC analyst:** You explored this career, where you respond to individual alerts and investigate them. The role might also be called cybersecurity analyst or triage analyst.
- **Threat hunter:** You explored this career, where you search for potential threats to a company or organization's computer systems and data.
- **Incident responder:** You explored the career of an incident responder, also known as an incident response analyst, who determines whether a reported alert is an organizational attack or a persistent threat to a company's network and coordinates a response.
- **Digital forensics investigator:** You explored this job, which retrieves, examines, and analyzes digital evidence to investigate cybercrimes.

Put your new skills and knowledge to use

You have new technical knowledge, stronger employability skills, and a better sense of your interest in the field of cybersecurity. There are many career paths you can take! You can get started on these paths by using your cybersecurity fluency to pursue a variety of jobs and contribute to any role or industry you work in.

Additional ideas for advancing your career

- Share your digital credentials (badges) on LinkedIn and other professional social media sites.
- Use the list of skills in this inventory to build your resume and prepare for job interviews.
- Reach out to a career counselor, mentor, or other ally for career opportunities and advice.
- Meet with a higher education professional to discuss options for further study.
- Create an elevator pitch that describes your career interests and your experience.
- Search for and attend networking events.
- Speak with your instructor to learn about other things you can do to progress your educational and career path.